Source: Sahil Bhola, Self-contained Proof for a Mixed-precision Block Jacobi Algorithm.

Selected deliverable: the full active result labeled theorem:block_dot_product in main.tex (lines 235-291).

The original mathematical proof is available in paper.tex. The selected statement below is authoritative.
Earlier statements and notation supply context. A referenced result may be proved as a helper. Do not assume a
desired algorithmic bound or an unproved paper lemma as an added hypothesis.

SELECTED STATEMENT (verbatim LaTeX, proof omitted)
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\begin{theorem}[Block Dot Product]\label{theorem:block_dot_product}
Let
\begin{align}
    y=\sum_{i=1}^{n}\vect{x}_i\vect{y}_i
\end{align}
be evaluated by partitioning the terms into
$n^\dagger=\ceil{n/T}$ blocks of size $T$, computing each block dot
product sequentially, and combining the resulting partial sums using a
binary-tree reduction of depth
$\ell^\dagger=\ceil{\log_2 n^\dagger}$. Any entries beyond $n$ in the
final block are taken to be zero. Assume that no overflow or underflow
occurs and that $(T+\ell^\dagger)\urd<1$. Then the computed result
$\hat{y}$ admits either of the backward-error representations
\begin{align}
    \hat{y}
    &=\sum_{i=1}^{n}
      \left(\vect{x}_i+\Delta\vect{x}_i\right)\vect{y}_i,
    &
    \abs{\Delta\vect{x}_i}
    &\leq \gamma_{T+\ell^\dagger}\abs{\vect{x}_i},
    \\
    \hat{y}
    &=\sum_{i=1}^{n}
      \vect{x}_i\left(\vect{y}_i+\Delta\vect{y}_i\right),
    &
    \abs{\Delta\vect{y}_i}
    &\leq \gamma_{T+\ell^\dagger}\abs{\vect{y}_i},
\end{align}
for all $i=1,\cdots,n$, where
$\gamma_{T+\ell^\dagger}
=(T+\ell^\dagger)\urd/(1-(T+\ell^\dagger)\urd)$.
\end{theorem}
```

SECTION CONTEXT
```
\section{Block Dot Product}
\label{sec:block_dot_product}
Another widely used kernel is computing the dot product kernel
\begin{align}
    y = \sum_{i=1}^n \vect{x}_i \vect{y}_i,
\end{align}
in a block-wise manner given by
\begin{align}
    y = \sum_{j = 1}^{n^\dagger}\sum_{t=1}^{T} \vect{x}_{(j-1)T + t} \vect{y}_{(j-1)T + t} =
\sum_{j=1}^{n^\dagger} p_j, \quad p_j \define \sum_{t=1}^{T} \vect{x}_{(j-1)T + t} \vect{y}_{(j-1)T + t},
\end{align}
where $n^\dagger \define \lceil n / T \rceil$ denotes the number of blocks of size $T$.
Typically, in parallel computing architecture like SIMT, the partial dot-product computation $p_j$ is computed
by independent threads and is computed sequentially.
Followed by, to reduce the partials $p_j$, we can perform computations such as sequential summation or parallel
binary tree reduction of depth $\log_2 n^\dagger$.
We first perform backward error analysis where the paritals $p_j$ are reduced using parallel binary tree
reduction.
Followed by, we also consider the case of sequential accumulation.
```

EARLIER STATEMENTS FOR CONTEXT
```
\begin{theorem}
\label{theorem:parallel_binary_tree_reduction}
Let
\begin{align}
    y=\sum_{i=1}^{n}\vect{x}_i
\end{align}
be evaluated componentwise using a binary-tree reduction of depth
$\ell^\dagger=\ceil{\log_2 n}$ in floating-point arithmetic with unit
roundoff $\urd$. Assume that no overflow or underflow occurs and that
$\ell^\dagger\urd<1$. Then the computed result $\hat{y}$ satisfies
\begin{align}
    \hat{y}
    =\sum_{i=1}^{n}\left(\vect{x}_i+\Delta\vect{x}_i\right),
    \qquad
    \abs{\Delta\vect{x}_i}
    \leq \gamma_{\ell^\dagger}\abs{\vect{x}_i},
    \quad i=1,\ldots,n,
\end{align}
where the inequalities are interpreted componentwise and
$\gamma_{\ell^\dagger}=\ell^\dagger\urd/(1-\ell^\dagger\urd)$.
\end{theorem}
```

COMMON ROUNDING MODEL AND ERROR CONSTANTS

```
\section{Floating-Point Model}
We assume the standard floating-point model~\cite{higham2002accuracy}: for every elementary
operation $\circ \in \{+,-,\times,/\}$,
\begin{equation}
    \fl(a \circ b) = (a \circ b)(1+\delta),
    \qquad
    \abs{\delta} \le \urd,
    \label{eq:fp-model}
\end{equation}
where $\urd$ is the unit roundoff and $\delta$ is the relative roundoff error.
Using~\cref{eq:fp-model} for numerical analysis we typicall obtain the product
\begin{align}
    \prod_{i=1}^n (1 + \delta_i)^{\rho_i} \define 1 + \theta_n, \qquad \rho = \pm 1,
    \label{eq:theta_n}
\end{align}
where $n$ denotes the number of accumulated roundoff error terms.
For numerical analysis, we seek reliable and sharp bounds for $\theta_n$.


\section{Deterministic Rounding Error Analysis}
The deterministic rounding error analysis proposes the following problem-size dependent constant
\begin{align}
\gamma_{n} \define \frac{n\urd}{1 - n\urd},
\label{eq:drea}
\end{align}
for $n\urd < 1$.
Here, $n$ is the number of accumulated error terms~\cite{higham2002accuracy}.
The problem-size dependent constant is monotonic, that is, $\gamma_{n+1} > \gamma_{n}$ for $\urd > 0$.
Performing backward/forward error analysis using~\cref{eq:drea} we obtain
\begin{align}
    \abs{\theta_n} \leq \gamma_n.
\end{align}

SOURCE NOTATION MACROS
\newcommand{\urd}{\ensuremath{\mathrm{u}}}
\newcommand{\define}{\ensuremath{\triangleq}}
\newcommand{\vect}[1]{\mathbf{\MakeLowercase{#1}}}
\newcommand{\mat}[1]{\mathbf{\MakeUppercase{#1}}}
\newcommand{\ireal}[1]{\in\mathbb{R}^{#1}}
\newcommand{\norm}[2][]{\left\|#2\right\|_{#1}}
\DeclareMathOperator{\fl}{fl}
\newcommand{\R}{\mathbb{R}}
\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\ceil}[1]{\left\lceil #1 \right\rceil}
\newcommand{\widehatgamma}[1]{\widehat{\gamma}_{#1}}
```